



Fully coupled multiphysics modeling of enhanced thermal conductivity $\text{UO}_2\text{--BeO}$ fuel performance in a light water reactor



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HIGHLIGHTS

- LWR fuel performance modeling capability developed.
- Fully coupled multiphysics studies for enhanced thermal conductivity $\text{UO}_2\text{--BeO}$ fuel.
- $\text{UO}_2\text{--BeO}$ fuel decreases fuel temperature and lessens thermal stresses.
- $\text{UO}_2\text{--BeO}$ fuel facilitates a reduction in PCMI.
- Reactor safety can be improved for $\text{UO}_2\text{--BeO}$ fuel.

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ABSTRACT

Commercial light water reactor fuel UO_2 has a low thermal conductivity that leads to the development of a large temperature gradient across the fuel pellet, limiting the reactor operational performance due to the effects that include thermal stresses causing pellet cladding interaction and the release of fission product gases. This study presents the development of a modeling and simulation for enhanced thermal conductivity $\text{UO}_2\text{--BeO}$ fuel behavior in a light water reactor, using self-defined multiple physics models fully coupled based on the framework of COMSOL Multiphysics. Almost all the related physical models are considered, including heat generation and conduction, species diffusion, thermomechanics (thermal expansion, elastic strain, densification, and fission product swelling strain), grain growth, fission gas production and release, gap heat transfer, mechanical contact, gap/plenum pressure with plenum volume, cladding thermal and irradiation creep and oxidation. All the phenomenal models and materials properties are implemented into COMSOL Multiphysics finite-element platform with a 2D axisymmetric geometry of a fuel pellet and cladding. $\text{UO}_2\text{--BeO}$ enhanced thermal conductivity nuclear fuel would decrease fuel temperatures and facilitate a reduction in pellet cladding interaction from our simulation results through lessening thermal stresses that result in fuel cracking, relocation, and swelling, so that the safety of the reactor would be improved.

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1. Introduction

Commercial light-water reactors currently use UO_2 as a nuclear fuel, which has demonstrated several desirable characteristics. However, UO_2 has a low thermal conductivity that leads to the development of a large temperature gradient across the fuel pellet. This temperature gradient contributes to limits in the operational performance of the reactor due to effects that include thermal stresses causing pellet cladding interaction and the release of fission product gases as demonstrated (Bailly et al., 1999; Belle, 1961; Frost, 1982; Holden, 1966). These high fuel temperatures

can be decreased and reactor performance improved by developing an enhanced thermal conductivity nuclear fuel. A high thermal conductivity nuclear fuel would decrease fuel temperatures and facilitate a reduction in pellet cladding interaction through lessening thermal stresses that result in fuel cracking, relocation, and swelling as described (Bailly et al., 1999; Frost, 1982). Additionally, fission gas release would be decreased allowing for higher fuel burn-up, and the safety of the reactor would be improved with a faster thermal response and less stored energy in the fuel pins. A ceramic–ceramic composite nuclear fuel has an increased effective thermal conductivity due to adding a second compatible and high thermal conductivity ceramic to UO_2 . SiC and BeO are two materials that demonstrate compatibility with UO_2 and have a high thermal conductivity as described (Kuchibhotla, 2004; Slack, 1973).

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Nomenclature

bu	burnup (%)
Bu	burnup (fissions/atoms-U)
C	atomic fraction of oxygen interstitials (atoms/m ³)
C	concentration of fission gas atoms (atoms/m ³)
C_p	specific heat (J/(kg K))
C_r	roughness coefficient
C_s	constant
C_1	rate constant for pre-transition oxidation (m/s)
C_2	rate constant for post-transition oxidation (m/s)
D	diffusion coefficient of fission gas (m ² /s)
D_f	fuel density in percent of theoretical one
d_g	gap width (m)
D_O	diffusion coefficient of oxygen (m ² /s)
d_{pen}	penetration distance (m)
E	Young's modulus (Pa)
E_D	activation energy for Frenkel defects (J/mole)
F	thermodynamic factor of oxygen
$\dot{F}$	volumetric fission rate (fissions/(m ³ s))
f_B	grain boundary fraction at saturation
f_d	dissolved fission products correction
f_p	precipitated fission products correction
f_{por}	porosity correction
f_r	radiation damage correction
f_x	deviation from stoichiometry
F_{dens}	fraction of fuel porosity removed by densification
F_v	body force per unit volume (N/m ³)
g	surface jump distance (m)
G	grain size (μ m)
G	shear modulus (Pa)
G_{bsat}	saturation fission gas release of Grain boundary (atoms/m ³)
g_d	average fuel grain diameter (m)
g_r	local fuel grain radius (m)
g_{max}	limiting grain size (m)
h	conductance (W/(m ² K))
H	Meyer hardness of the softer material (Pa)
h_{film}	cladding effective heat transfer coefficient (W/(m ² K))
J	oxygen flux (atoms/(m ² s))
k	rate constant (m ² /s)
k	thermal conductivity (W/(m K))
k_{95}	thermal conductivity of 95%TD fuel (W/(m K))
k_B	Boltzmann constant (m ² /(s ² K))
n	moles of gas moles
p	fuel porosity fraction
P	pressure (Pa)
P_c	contact pressure (Pa)
P_{est}	estimated contact pressure (Pa)
P_{ext}	externally applied hydrostatic pressure (Pa)
P_f	penalty factor (Pa/m)
P_i	interface pressure (Pa)
P_{lin}	linear power (W/m)
p_0	fuel initial porosity fraction
Q	heat generation rate (W/m ³)
Q_{fg}	volumetric rate of production of fission gas atoms (atoms/(m ³ s))
Q_i	activation energies (J/mol)
Q_r	net radial heat flux from cladding to bulk coolant (W/m ²)
Q_{oxide}	heat flux reduction due to cladding oxide layer growth (W/m ²)

Q_1	activation energy for pre-transition oxidation (J/mol)
Q_2	activation energy for post-transition oxidation (J/mol)
R	universal gas constant (J/(mol K))
r	surface roughness (m)
r_f	grain boundary bubble radius (m)
R_{gb}	fission gas release rate to grain boundary (atoms/(m ³ m s))
T	temperature (K)
T_n	$T_n = T(K)/1000$
t	time (s)
t_{oxide}	oxide layer thickness (m)
t_{trans}	oxide layer thickness at transition period (m)
$\mathbf{u}$	displacement vector (m)
V	volumetric fraction
V	volume (m ³)
W	weight fraction
Y	oxygen to metal ratio
X_{dev}	oxygen to metal ratio deviation
α	coefficient of thermal expansion
a_{pellet}	fuel pellet radius (m)
δ	average gas film thickness (m)
σ	Stephan-Boltzman constant (kg/(s ³ K ⁴))
σ	Cauchy stress tensor (Pa)
σ	effective (Mises) stress (Pa)
ε	radiating surface emissivity
ε	elastic strain vector
$\dot{\varepsilon}$	creep rate (1/s)
$\dot{\varepsilon}_{ir}$	effective irradiation creep rate (1/s)
$\dot{\varepsilon}_{ss}$	effective thermal creep rate (1/s)
ε_D	fraction of volume reduction due to fuel densification
ρ	density (kg/m ³)
θ	the Einstein temperature (K)
θ_{fg}	semi-dihedral angle between bubble surfaces (°)
ν	Poisson's ratio
Φ	fast neutron flux (n/(m ² s))
γ_{se}	surface tension of the bubbles (J/m ²)
$\Delta\varepsilon_{sw-s}$	volumetric solid swelling increment (m ³)
$\Delta\varepsilon_{sw-g}$	volumetric gas swelling increment (m ³)
FEA	finite element analysis
FGR	fission gas release
LWR	light water reactor

Understanding and predicting the evolving behaviors of nuclear fuels are major challenges for enhanced thermal conductivity fuels' design, performance, and safe operation. The multiphysics behavior of nuclear fuels is often a tightly coupled thermal, mechanical and chemical phenomenon associated with a large number of control parameters and the uncertainties, which requires advanced fuel performance modeling and simulation capability to improve reactor fuel performance, to address environmental and reactor safety concerns.

Nuclear fuel performance code is necessary as a result of the high cost and difficulty in nuclear experimental measurements. Previous modeling codes employed either a fixed one-dimensional or a quasi-two-dimensional representation of the fuel geometry such as GAPCON (Hann et al., 1973), FALCON (Rashid et al., 2004), FEMAXI (Nakajima et al., 1994), FRAPCON (Berna et al., 1997) and FRAPTRAN (Geelhood et al., 2011). These codes typically utilize many empirical correlations with limited coupling between phenomena. More notable codes like French codes TOUTATIS

(Bentejac and Hourdequin, 2004) and ALCYONE (Thouvenin et al., 2007) are three-dimensional and used to investigate localized behavior. However, these previous representative codes are all written in FORTRAN, which is developed as purpose-written, standalone computer programs. The main disadvantage of this architecture is that significant modifications are difficult, and generally require editing the source code.

Recently, the BISON code (Williamson et al., 2012) is developed by Idaho National Laboratory employing a full set of multidimensional multiphysics capabilities, which can be used to solve 1D, 2D axisymmetric and 3D nuclear fuel performance problems and can also solve all unknowns (displacements, stresses, temperature, fission gas release, plenum pressure, etc.) concurrently at each time. This code is not an open source and also needs a long time to develop, and one has to have intimate knowledge of the internal structure of the code for modeling different fuels and/or geometries. Williamson (2011) also used commercial software ABAQUS to enhance to modeling of multipellet steady and transient LWR fuel rod behavior.

More recently, an extremely flexible FAST code is developed by using commercial COMSOL Multiphysics to simulate the CANDU reactor fuel performance under normal and transient conditions (Prudil et al., 2015a,b). It assumes azimuthal symmetry allowing the use of a two-dimensional (radial-axial) representation of the fuel element geometry including pellet dishing, chamfering and central holes. It also assumes that the single pellet is representative of all pellets within an element with considering a periodic boundary condition in the axial direction. The sheath is also represented in 2D (Prudil et al., 2015a,b). This code includes models solved for temperature, displacement, grain size, burnup, densification, swelling, fission gas diffusion and release to grain boundary, the pellet-to-pellet gap size, pellet-to-sheath gap size and sheath creep. It presents proof-of-concept comparison to experimental data and its predictions were verified against those of the ELESIM (Tayal, 1987) and ELESTRES (AECL, 1975) codes for CANDU fuel performance.

In this paper, a CityU Advanced Multiphysics Nuclear Fuels Performance with User-defined Simulations code (CAMPUS) has been developed. CAMPUS is based on the framework of COMSOL Multiphysics and uses user-defined multiple physical models. COMSOL Multiphysics is a finite element analysis, solver and simulation software/FEA software package for various physics and engineering applications, especially coupled phenomena, or multiphysics (COMSOL, Inc., 2015). The script language is MATLAB for COMSOL Multiphysics. The CAMPUS modeling and simulation code structure is shown in Fig. 1. COMSOL Multiphysics framework is used to develop CAMPUS fuel performance code and predict the fuel behavior. In the approach, the materials properties and performance were analyzed, modeled, and implemented into the code. These multiple physics are nonlinear with nonlinear material properties and strongly coupled. CAMPUS was then used to model and predict $\text{UO}_2\text{-BeO}$ fuel performance with different BeO fractions in a light water reactor. The simulation geometry includes fuel pellet, gap, plenum and cladding. The materials properties are more accurate parameters using temperature dependent, burnup dependent or other function based parameters other than constant values used in some previous approaches. Almost all the physical models were implemented into the simulation in a fully coupled way. Through fully coupled multiphysics modeling, a more complete and detailed understanding was achieved for enhanced thermal conductivity $\text{UO}_2\text{-BeO}$ fuel performance in a light water reactor, which has never been done before according to the authors' knowledge.

2. Materials properties

2.1. $\text{UO}_2\text{-BeO}$ fuel properties

(1) Thermal conductivity

In general, the $\text{UO}_2\text{-BeO}$ fuel thermal conductivity is given as in Eq. (1) accounting for the effects of dissolved fission products, precipitated fission products, porosity, deviation from stoichiometry,

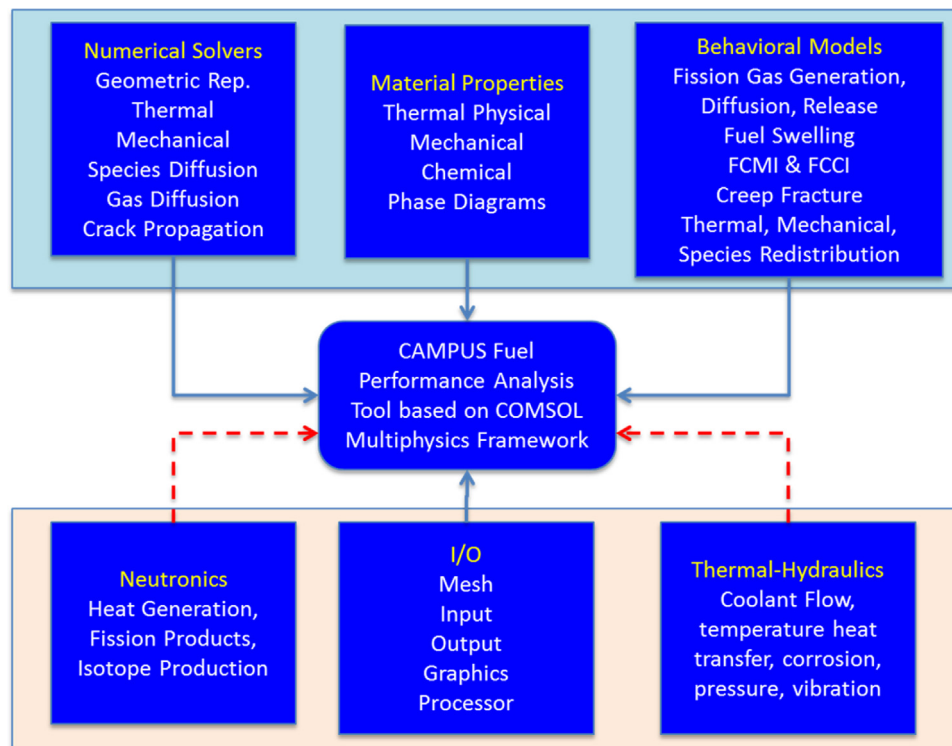


Fig. 1. CAMPUS fuel performance modeling and simulation structure.

Table 1
 UO₂–BeO measured thermal conductivities (Latta et al., 2008).

BeO 0.6 wt%		BeO 1.2 wt%		BeO 13.6 wt%	
T (K)	k (W/mK)	T (K)	k (W/mK)	T (K)	k (W/mK)
479	9.4	479	10.3	500	38.0
688	6.8	688	7.6	700	25.0
903	5.4	903	5.6	900	17.5
1097	4.6	1097	4.5	1100	12.5
1278	3.8	1278	3.9	1300	11.0
1486	3.3	1486	3.3	1500	9.0
1653	3.0	1653	3.2		
1860	2.8	1860	2.8		

and radiation damage, as for UO₂ fuel (Lucuta et al., 1996; Fink, 2000)

$$k = k(\text{UO}_2\text{--BeO}) \cdot f_d \cdot f_p \cdot f_{por} \cdot f_x \cdot f_r \quad (1)$$

where f_d – dissolved fission products correction; f_p – precipitated fission products correction; f_{por} – porosity correction; f_x – deviation from stoichiometry; f_r – radiation damage correction.

The as-fabricated UO₂–BeO fuel thermal conductivity is calculated by multiplying a factor to 95%TD UO₂–BeO fuel thermal conductivity as shown in Eq. (2):

$$k(\text{UO}_2\text{--BeO}) = k_{95}(\text{UO}_2\text{--BeO}) \left(\frac{1}{1 - (2.6 - 0.5T_n) \cdot 0.05} \right) \quad (2)$$

The temperature dependent thermal conductivity of unirradiated 95%TD UO₂–BeO fuel is defined using Eq. (3):

$$k_{95}(\text{UO}_2\text{--BeO}) = \frac{1}{A + BT} + \frac{E}{T^2} \exp\left(-\frac{F}{T}\right) \quad (3)$$

where T – temperature.

The constants in Eq. (3) are determined by curve fitting using the measured UO₂–BeO fuel thermal conductivities from Latta et al. (2008). The measured thermal conductivities and the constants determined are listed in Tables 1 and 2, respectively. The constants for UO₂ fuel, i.e., 0 wt% BeO, are from Geelhood et al. (2011).

The relationship between the weight fraction and volumetric fraction is as following:

$$\begin{cases} W_{\text{UO}_2} = \frac{V_{\text{UO}_2} \cdot \rho_{\text{UO}_2}}{V_{\text{UO}_2} \cdot \rho_{\text{UO}_2} + V_{\text{BeO}} \cdot \rho_{\text{BeO}}} \\ W_{\text{BeO}} = \frac{V_{\text{BeO}} \cdot \rho_{\text{BeO}}}{V_{\text{UO}_2} \cdot \rho_{\text{UO}_2} + V_{\text{BeO}} \cdot \rho_{\text{BeO}}} \end{cases} \quad (4)$$

The comparison is shown in Fig. 2 between the measured thermal conductivities and the ones determined by Eq. (3) using the constants from Table 2. The thermal conductivity of BeO is also shown in Fig. 2 as a reference, which is taken from Touloukian et al. (1970).

The correction factors in Eq. (1) are defined as prescribed by Lucuta et al. (1996) where

$$f_d = \left(\frac{1.09}{bu^{3.265}} + 0.0643 \cdot \sqrt{\frac{T}{bu}} \right) \cdot \arctan \left(\frac{1.0}{\frac{1.09}{bu^{3.265}} + 0.0643 \cdot \sqrt{\frac{T}{bu}}} \right) \quad (5)$$

Table 2
 Constants used in Eq. (3).

BeO		A	B	E	F
wt%	vol%	m K/W	m/W	m K/W	K
0	0	0.0452	2.46e–4	3.5e9	16,361
0.6	2.14558	0.01752	1.92e–4	3.5e9	16,361
1.2	4.22546	0.007068	1.95e–4	3.5e9	16,361
13.6	36.3777	–0.01378	8.21e–5	3.5e9	16,361

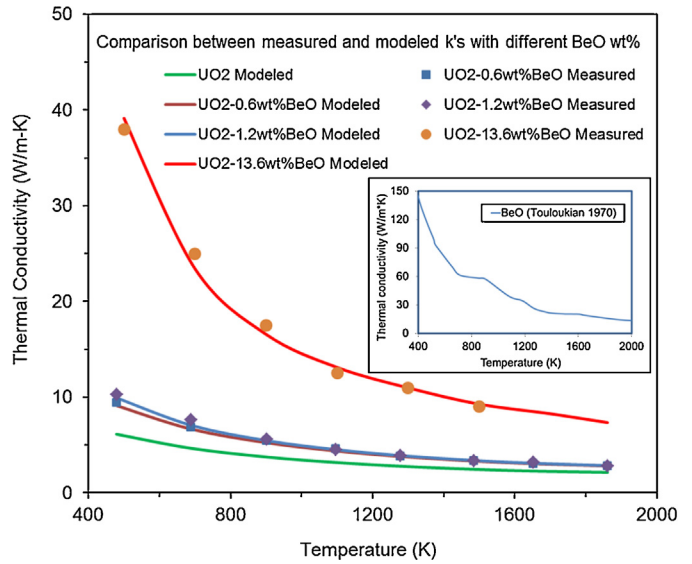


Fig. 2. Thermal conductivity of UO₂ and UO₂–BeO fuels from measured and modeled results.

$$f_p = 1.0 + \left(\frac{0.019 \cdot bu}{3.0 - 0.019 \cdot bu} \right) \cdot \left(\frac{1.0}{1.0 + \exp\left(-\frac{(T-1200)}{100}\right)} \right) \quad (6)$$

$$f_{por} = \left(\frac{1.0 - p}{1.0 + 0.5 \cdot p} \right) \quad (7)$$

$$f_r = 1.0 - \frac{0.2}{1.0 + \exp\left(-\frac{(T-900)}{80}\right)} \quad (8)$$

where T is the temperature in K, p is the porosity, and bu is the burnup in at%. The Fink–Lucuta model is valid from 298 to 3120 K (Lucuta et al., 1996; Fink, 2000).

(2) Heat capacity

The heat capacity of composite is a combination of weight percentage multiplied by single component heat capacity for both UO₂ and BeO.

$$C_p(\text{UO}_2\text{--BeO}) = W_{\text{UO}_2} \cdot C_p(\text{UO}_2) + W_{\text{BeO}} \cdot C_p(\text{BeO}) \quad (9)$$

The MATPRO-11 correlation for the heat capacity of Urania was used in the simulation (Carbajo et al., 2001):

$$C_p(\text{UO}_2) = \frac{K_1 \theta^2 \exp(\theta/T)}{T^2 [\exp(\theta/T) - 1]^2} + K_2 T + \frac{YK_3 E_D}{2RT^2} \exp(-E_D/RT) \quad (10)$$

where T – temperature; K_1 – constant, 296.7 J/kg K; K_2 – constant, 0.0243 J/kg K²; K_3 – constant, 8.75e7 J/kg; Y – oxygen to metal ratio; R = universal gas constant = 8.315 J/mole K; θ = Einstein temperature (K), for Urania = 535.285 K; E_D = activation energy for Frenkel defects (J/mole), for Urania = 1.577e5 J/mole.

The correlation for the heat capacity of BeO from Chandramouli and Revankar (2014) was used in the simulation.

$$C_p(\text{BeO}) = 1000 \cdot (0.036 \cdot ((T - 650)/360)^3 - 0.12 \cdot ((T - 650)/360)^2 + 0.2 \cdot ((T - 650)/360) + 1.9) \quad (11)$$

The heat capacities of UO₂, BeO and UO₂–BeO are shown in Fig. 3 assuming O/M ratio = 2.

(3) Density

UO₂–BeO fuel density is calculated using a combination of volumetric fraction multiplied by single component density for both UO₂ and BeO:

$$\rho(\text{UO}_2\text{--BeO}) = V_{\text{UO}_2} \cdot \rho(\text{UO}_2) + V_{\text{BeO}} \cdot \rho(\text{BeO}) \quad (12)$$

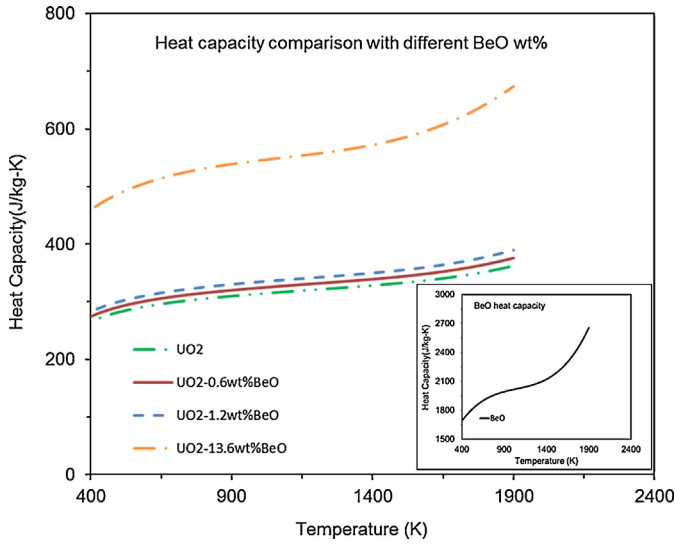


Fig. 3. Heat capacity of UO_2 , BeO and $\text{UO}_2\text{-BeO}$ assuming O/M ratio = 2.

The density of solid UO_2 recommended by Fink (2000) was used in the simulation:

$$\rho(\text{UO}_2) = 10,970(a + bT + cT^2 + dT^3)^{-3}, \text{ with}$$

For $T \leq 923 \text{ K}$:

$$a = 0.99734, b = 9.802 \times 10^{-6}, c = -2.705 \times 10^{-10}, d = 4.391 \times 10^{-13} \quad (13)$$

For $T > 923 \text{ K}$:

$$a = 0.99672, b = 1.179 \times 10^{-5}, c = -2.429 \times 10^{-9}, d = 1.219 \times 10^{-12}$$

The density of solid BeO recommended by IAEA (2008) was used in the simulation:

$$\rho(\text{BeO}) = 0.21 \cdot ((T - 1200)/530)^4 + 2.6 \cdot ((T - 1200)/530)^3 - 3 \cdot ((T - 1200)/530)^2 - 63 \cdot ((T - 1200)/530) + 2900 \quad (14)$$

The densities of UO_2 , BeO and $\text{UO}_2\text{-BeO}$ are shown in Fig. 4.

(4) Young's modulus, Poisson's ratio and thermal expansion coefficient for $\text{UO}_2\text{-BeO}$ fuel used combinations of volumetric fractions multiplied by single component Young's modulus, Poisson's ratio and thermal expansion coefficient for both UO_2 and BeO, i.e.,

$$E(\text{UO}_2\text{-BeO}) = V_{\text{UO}_2} \cdot E(\text{UO}_2) + V_{\text{BeO}} \cdot E(\text{BeO}) \quad (15)$$

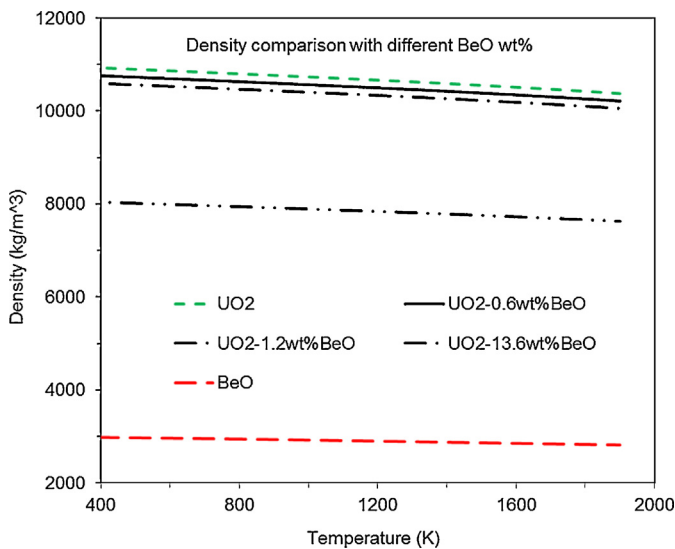


Fig. 4. Density of UO_2 , BeO and $\text{UO}_2\text{-BeO}$.

$$\nu(\text{UO}_2\text{-BeO}) = V_{\text{UO}_2} \cdot \nu(\text{UO}_2) + V_{\text{BeO}} \cdot \nu(\text{BeO}) \quad (16)$$

$$\frac{\Delta L}{L}(\text{UO}_2\text{-BeO}) = V_{\text{UO}_2} \cdot \frac{\Delta L}{L}(\text{UO}_2) + V_{\text{BeO}} \cdot \frac{\Delta L}{L}(\text{BeO}) \quad (17)$$

The thermal expansion of solid UO_2 recommended by Martin (1988) was used in the simulation:

$$\frac{\Delta L}{L} = a + bT + cT^2 + dT^3, \text{ with}$$

For $T = 923 \text{ K}$:

$$a = 9.828 \times 10^{-6}, b = -6.390 \times 10^{-10}, c = 1.33 \times 10^{-12}, d = -1.757 \times 10^{-17}$$

For $T > 923 \text{ K}$:

$$a = 1.183 \times 10^{-5}, b = -5.013 \times 10^{-9}, c = 3.756 \times 10^{-12}, d = -6.125 \times 10^{-17} \quad (18)$$

Young's modulus of solid UO_2 recommended by Martin (1988) was used in the simulation:

$$E = ((2.334e11) \cdot (1 - (1.0915e - 4) \cdot T \cdot \exp(-1.34X_{dev}))) \quad (19)$$

where X_{dev} – oxygen to metal ratio deviation. Poisson ratio of solid UO_2 recommended by Martin (1988) was used in the simulation:

$$\nu(\text{UO}_2) = 0.316 + (0.5 - 0.316)(T - 300)/2800 \quad (20)$$

The coefficient of thermal expansion of BeO is calculated as in IAEA (2008):

$$\alpha \times 10^6 (1/K) = 5.133 + 4.65 \times 10^{-3}T - 1.539 \times 10^{-7}T^2 - 3.621 \times 10^{-10}T^3 \quad (21)$$

The Young's modulus and Poisson's ratio of BeO were $3.5 \times 10^{11} \text{ Pa}$ and 0.229 (Soga, 1969), respectively, in the simulation.

(5) Oxygen diffusion coefficient is defined as below in the simulation (Lassmann, 1987):

$$D_0 = 1.39e - 6 \cdot \exp(-9128/T) \quad (22)$$

2.2. Zircaloy cladding properties

Zircaloy cladding properties were adopted from IAEA thermo-physical properties database report (IAEA, 2006 and EG&G Idaho, Inc., 1976).

(1) Cladding thermal conductivity

$$k_{Zr} = 11.498 + 4.6765 \times 10^{-3} \cdot T + 2.761 \times 10^{-6} \cdot T^2 + 2.2147 \times 10^{-9} \cdot T^3 \quad (23)$$

(2) Heat capacity

$$c_{p,Zr} = (255.66 + 0.1024T) \quad (24)$$

(3) Density

$$\rho_{Zr} = 6595.2 - 0.1477T \quad (25)$$

(4) Poisson ratio

$$\nu_{Zr} = 0.32 \quad (26)$$

(5) Young's modulus

$$E_{Zr} = 108.8 \times 10^9 - 5.475 \times 10^7 T \quad (27)$$

(6) Thermal expansion

$$\alpha_{Zr} = \begin{cases} 7.092 \times 10^{-6}, & \text{radial} \\ 9.999 \times 10^{-6}, & \text{angular} \\ 5.458 \times 10^{-6}, & \text{axial} \end{cases} \quad (28)$$

2.3. Gap properties

The gap between fuel and cladding is filled with helium gas, with the properties obtained from *Heat Transfer* by Bejan (1993).

(1) Helium density

$$\rho_{He} = 0.0818 - 8.275 \times 10^{-15}(T - 600) \quad (29)$$

(2) Helium heat capacity

$$c_{p,He} = 5190 \quad (30)$$

(3) Helium thermal conductivity

$$k_{He} = 0.04679 + 3.81 \times 10^{-4}T - 6.786 \times 10^{-8}T^2 \quad (31)$$

3. Models implementation

3.1. Model geometry

The model geometry in this work adopted a 2D axisymmetric plane with a Zr-4 cladding (see Fig. 5(a)). A single pellet was considered to represent all the pellets with a periodic boundary condition in the axial direction, as shown in Fig. 5(b), with a mapped mesh.

3.2. Heat generation and conduction

The commercially available COMSOL software provides a reasonable framework for fuel performance modeling. The heat transfer model was implemented using the “Heat transfer in Solids” physics module included in COMSOL. This module defines the temperature, T (K), by the heat conduction equation:

$$\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q \quad (32)$$

where ρ , C_p and k are the material properties of density, specific heat capacity and thermal conductivity, respectively. All the material properties were defined in the “Materials” module. The volumetric heating term, Q , accounts for heat generated in the fuel.

For single pellet case, a uniformly distributed heat source term were considered, which is given by:

$$Q = \frac{P_{lin}}{\pi \cdot a_{pellet}^2} \quad (33)$$

where P_{lin} is the linear power, a_{pellet} is fuel pellet radius.

Heat transport across the pellet-to-cladding gap is modeled assuming one-dimensional steady-state heat transfer by using the following radial heat flux:

$$Q_r = (h_{gas} + h_{solid} + h_{rad})(T_{fuel} - T_{cladding}) \quad (34)$$

where the total conductance across the gap is the sum of gas conductance (h_{gas}), conductance due to solid–solid contact (h_{solid}) and the conductance due to radiant heat transfer (h_{rad}). This model is typically applied between the fuels and clad. The gas conductance h_{gas} is described using the form proposed by Ross and Stoute (1962):

$$h_{gas} = \frac{k_{gas}(T_{gas})}{d_g + C_r(r_1 + r_2) + g_1 + g_2} \quad (35)$$

where k_{gas} is the conductivity of the gas in the gap, d_g is the gap width (computed in the mechanics solution), C_r is a roughness coefficient with r_1 and r_2 being the roughnesses of the two surfaces, and g_1 and g_2 are jump distances at the two surfaces.

The conductance due to solid–solid contact, h_{solid} , is given by the empirical model suggested by Olander (1976):

$$h_{solid} = C_s \frac{2k_1k_2}{k_1 + k_2} \frac{P_c}{\delta^{0.5}H} \quad (36)$$

where C_s is a constant (typically 1.0), k_1 and k_2 are the thermal conductivities of the solid materials in contact, P_c is the contact pressure which is modeled by using a penalty method, δ is the average gas film thickness, and H is the Meyer hardness of the softer material.

The conductance due to radiative heat transfer, h_{rad} , is calculated using a diffusion approximation. Based on the Stephan–Boltzmann law:

$$q_{rad} = \sigma \frac{1}{(1/\varepsilon_1) + (1/\varepsilon_2) - 1} (T_1^4 - T_2^4) \approx h_{rad}(T_1 - T_2) \quad (37)$$

where σ is the Stephan–Boltzman constant, ε_1 and ε_2 are the emissivity of the radiating surfaces, T_1 and T_2 are the temperatures of

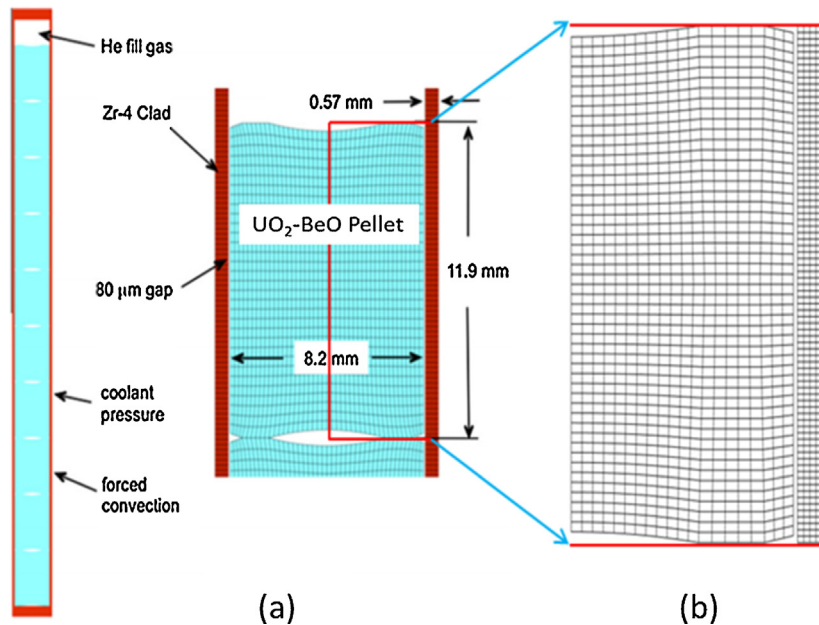


Fig. 5. (a) Geometry, material and typical mesh used in BISON, (b) 2D-axisymmetric one pellet geometry and mesh used in this work.

the fuel outer surface and cladding inner surface. The radiative conductance can be written in the form of following:

$$h_{rad} = \sigma \frac{1}{(1/\varepsilon_1) + (1/\varepsilon_2) - 1} (T_1^2 + T_2^2)(T_1 + T_2) \quad (38)$$

The heat conductance from the fuel cladding to the bulk coolant considered as thin-film heat transfer, the net radial heat flux from the cladding to the bulk coolant, Q_r , is calculated as:

$$Q_r = \frac{T_{cladding} - T_{coolant}}{((t_{oxide}/k_{oxide}) + (1/h_{film}))} - Q_{oxide} \quad (39)$$

where $T_{cladding}$ is cladding temperature and $T_{coolant}$ is coolant temperature. k_{oxide} is the thermal conductivity of zirconium-oxide layer and t_{oxide} is the layer thickness. h_{film} is the Zircaloy-4 cladding effective heat transfer coefficient, Q_{oxide} is reduction of heat flux from the fuel cladding due to the growth of the oxide layer and is given by Olander (1976):

$$Q_{oxide} = \frac{42.426 \times 10^9}{1.56} \frac{dt_{oxide}}{dt} \quad (40)$$

where zirconium-oxide layer thickness t_{oxide} is based on the corrosion rate. For the pre-transition period, the corrosion rate is given by an Arrhenius equation (Leistikow and Schanz, 1987):

$$\frac{dt_{oxide}^3}{dt} = C_1 \exp\left(\frac{-Q_1}{RT_i}\right) \text{ for } t_{oxide} \leq t_{trans} \quad (41)$$

For the post-transition period, the corrosion rate is given by an Arrhenius equation (Ritchie, 1998):

$$\frac{dt_{oxide}}{dt} = C_2 \exp\left(\frac{-Q_2}{RT_i}\right) \text{ for } t_{oxide} > t_{trans} \quad (42)$$

where C_1 , C_2 and Q_1 , Q_2 are the rate constants and activation energy for pre-transition oxidation and post-transition oxidation respectively.

3.3. Oxygen diffusion model

A hyperstoichiometric model for oxygen diffusion is adopted from Newman et al. (2009). In this case J denotes the oxygen flux in the hyperstoichiometric regime with,

$$J = -D_o \left(\nabla C + \frac{CQ^*}{FRT^2} \nabla T \right) \quad (43)$$

where D_o is the oxygen diffusivity, F is the thermodynamic factor of oxygen, C is the atomic fraction of oxygen interstitials and R is the universal gas constant. The oxygen flux J satisfies the equation of continuity expressing the oxygen atoms conservation:

$$\frac{\partial C}{\partial t} + \nabla \cdot J = 0 \quad (44)$$

3.4. Deformation mechanics

3.4.1. Elastic deformation

In the reactor, the fuel element deformation behavior is described by using Cauchy's equation, which can be found in the COMSOL structural mechanics module given as the following form:

$$-\nabla \cdot \sigma = F_v \quad (45)$$

where σ is the Cauchy stress tensor and F_v is the body force per unit volume, which can be forms including: applied forces, thermal expansion, materials creep, fuel densification and fission gas swelling. The stress is calculated with a liner elastic constitutive model:

$$\sigma = [C][\varepsilon] \quad (46)$$

where C is the material matrix, ε is the elastic strain vector defined as $1/2[\nabla \cdot u + \nabla \cdot u^T]$, and u is the displacement vector. The total strain is calculated in terms of the spatial derivatives of the displacements. In the 2D axisymmetric case, as used in this paper, the strain components are given by:

$$\begin{aligned} \varepsilon_r &= \frac{\partial u}{\partial r}, \quad \varepsilon_\theta = \frac{u}{R}, \quad \varepsilon_z = \frac{\partial v}{\partial z}, \\ \varepsilon_{rz} &= \frac{1}{2} \left(\frac{\partial v}{\partial r} + \frac{\partial u}{\partial z} \right), \quad \varepsilon_{r\theta} = \varepsilon_{\theta z} = 0. \end{aligned} \quad (47)$$

The penalty method from FAST code (Prudil et al., 2015a,b) was used to model the pellet-to-cladding contact with a force applied normal to the cladding. As stated clearly in FAST code (Prudil et al., 2015a,b), the penalty function was selected such that the interface pressure, P_i , increases exponentially to the estimated contact pressure (P_{est}) as the fuel and cladding boundaries come into contact and then increases linearly with the penetration distance (d_{pen}) at a rate determined by the penalty factor, P_f , to prevent extremely high stiffness which could occur if the exponential term was still used (Prudil et al., 2015a,b).

$$P_i = \begin{cases} P_{est} \exp\left(\frac{P_f d_{pen}}{P_{est}}\right) & d_{pen} < 0 \\ P_{est} + P_f d_{pen} & d_{pen} \geq 0 \end{cases} \quad (48)$$

Both the estimated contact pressure and the penalty factor are adjustable parameters. For the results of light water reactor case presented in this work, values of $P_{est} = 1$ MPa and $P_f = 5 \times 10^{12}$ Pa/m were used for the pellet-to-cladding contact. The pellet-to-pellet contact is neglected in this work as we considered a virtual fuel pellet. The friction between the contact surfaces has also been neglected in this work as its small expected contribution and significant increase of the computation complexity.

3.4.2. Thermal strain/expansion

The thermal expansion model in Eq. (15) was used in this work considering an isotropic UO_2 -BeO fuel pellet.

3.4.3. Fuel densification and fission product swelling

Fuel densification is computed by using the empirical correlation developed by Hastings and Evans (1979):

$$F_{dens} = 0.6 - \exp(-0.506 - 8.67 \times 10^{-10} T^3 (1 - \exp(-2.867 \times 10^{-2} Bu))) \quad (49)$$

$$\varepsilon_D = \frac{\Delta V_{dens}}{V} = \left(\frac{1 - p_0}{1 - p_0(1 - F_{dens})} - 1 \right) \quad (50)$$

where F_{dens} is the fraction of the fuel porosity which is removed by densification, p_0 is the initial porosity.

Swelling due to both solid and gaseous fission products is considered using empirical relations from MATPRO (Allison et al., 1993). The volumetric strain due to solid fission product swelling is given as a simple linear function of burnup:

$$\Delta \varepsilon_{sw-s} = 5.577 \times 10^{-5} \rho \Delta Bu \quad (51)$$

where $\Delta \varepsilon_{sw-s}$ is the volumetric solid swelling increment, ΔBu the burnup increment (fissions/atoms-U), and ρ is the density (kg/m^3). Swelling due to gaseous fission products is calculated by a semi-empirical model from MATPRO (Allison et al., 1993):

$$\begin{aligned} \Delta \varepsilon_{sw-g} &= 1.96 \times 10^{-31} \rho \Delta Bu (2800 - T)^{11.73} \\ &\times e^{-0.0162(2800-T)} e^{-0.0178 \rho Bu} \end{aligned} \quad (52)$$

where $\Delta\epsilon_{sw-g}$ is the volumetric gas swelling increment, Bu and ΔBu are the burnup and burnup increment (fissions/atoms-U), respectively, ρ is the fuel density (kg/m³) and T is the fuel temperature (K).

3.4.4. Fuel thermal and irradiation creep

A MATPRO FCREEP material model (Allison et al., 1993) for combined secondary thermal creep and irradiation creep of UO₂ is used in this work, given as:

$$\dot{\epsilon} = \frac{A_1 + A_2 \dot{F}}{(A_3 + D_f)G^2} \sigma e^{(-Q_1/RT)} + \frac{A_4}{(A_6 + D_f)} \sigma^{4.5} e^{(-Q_2/RT)} + A_7 \dot{F} \sigma e^{(-Q_3/RT)} \quad (53)$$

where $\dot{\epsilon}$ is the creep rate (1/s), σ is the effective (Mises) stress (Pa), T is the temperature (K), D_f is the fuel density (in percent of theoretical), G is the grain size (μ m), $\dot{F}$ is the volumetric fission rate (fissions/m³/s), Q_i are the activation energies (J/mol), R is the universal gas constant and A_{1-7} are material constants.

3.4.5. Cladding thermal and irradiation creep

Zircaloy was considered as the cladding, and secondary thermal creep model recommended by Hayes and Kassner (2006) was used as shown below:

$$\dot{\epsilon}_{ss} = A_0 \left(\frac{\sigma_m}{G} \right)^n \exp \left(\frac{-Q}{RT} \right) \quad (54)$$

where $\dot{\epsilon}_{ss}$ is the effective thermal creep rate (1/s), σ_m is the effective (Mises) stress (Pa), Q is the activation energy (J/mol), R is the universal gas constant (J/(mol K)), T is the temperature (K), G is the shear modulus (Pa) recommended by Moon et al. (2006) as $G = 4.2519 \times 10^{10} - 2.2185 \times 10^7 T$, n is specified by Hayes and Kassner (2006) as a power of 5, and A_0 is with a value of 3.14×10^{24} (1/s). Irradiation-induced cladding creep is implemented from Hooke (1991), which relates the creep rate to the current fast neutron flux and stress:

$$\dot{\epsilon}_{ir} = C_0 \Phi^{C_1} \sigma_m^{C_2} \quad (55)$$

where $\dot{\epsilon}_{ir}$ is the effective irradiation creep rate (1/s), Φ is the fast neutron flux (n/m^2 s), σ_m is the effective stress (Pa), and C_0 , C_1 and C_2 are material constants. An effective (Mises) stress was used rather than the original Hooke's circumferential stress formulation.

3.5. Fission gas production and release

The fission gas release (FGR) was calculated by using the Forsberg–Massih two-stage model (Forsberg and Massih, 1985), namely, in the first stage, the fission gas is produced in the fuel grains and diffuses to the fuel grain boundary and then forms intergranular lenticular bubbles. The second stage occurs when the fuel grain surface gets saturated and any additional fission gas at the grain surface is released to the free volume of the element.

3.5.1. Fission gas release to the grain boundaries

Booth diffusion model (White and Tucker, 1983) was implemented into our simulation, which is described in the following equation form:

$$\frac{\partial C}{\partial t} = -D \nabla^2 C + Q_{fg} \quad (56)$$

where C is the concentration of fission gas atoms in the fuel grains, D is the diffusion coefficient of fission gas in the fuel grain and Q_{fg} is the volumetric rate of production of fission gas atoms. A two-dimensional rectangular domain is used in which one direction

represents the radial coordinate of the grain within the fuel pellet and the other direction represents the radial coordinate within the spherical grain. Eq. (56) was transformed to a non-dimensional form by substitution of $\eta = \frac{r}{g_r}$ to obtain

$$\eta^2 g_r^2 \frac{\partial C}{\partial t} = \frac{\partial C}{\partial \eta} \left(-D \eta^2 \frac{\partial C}{\partial \eta} \right) + \eta^2 g_r^2 Q_{fg} \quad (57)$$

Then the release rate to the grain boundary, R_{gb} , is

$$R_{gb} = \frac{3}{g_r^2} D \frac{\partial C}{\partial \eta} \bigg|_{\eta=1} \quad (58)$$

where, g_r is the local fuel grain radius, the local grain size was determined by a simple empirical model (Ainscough et al., 1973). The kinetics of grain growth is described by the equation:

$$\frac{dg_d}{dt} = k \left(\frac{1}{g_d} - \frac{1}{g_{max}} \right) \quad (59)$$

where g_d is the average grain diameter ($2g_r$), t (s) the time, k (m²/s) the rate constant, which is given by $1.46 \times 10^{-8} \exp(-32,100/T)$, T is the temperature in K, g_{max} is the limiting grain size (m) as a function of temperature, which is

$$g_{max} = 2.23 \times 10^{-3} \exp \left(\frac{-7620}{T} \right) \quad (60)$$

3.5.2. Fission gas release to the fuel element

After the fission gas has been released to the grain surface, it will be trapped in the intergranular bubbles among fuel grains. Once the gases were accumulated on grain boundaries become saturated (the saturated gas concentration G_{bsat}), the fission gas will release to the fuel free volume. The grain boundary saturation, G_{bsat} (atoms/m³), is given by White and Tucker (1983) as

$$G_{bsat} = \frac{4r_f f(\theta_{fg}) f_B}{3k_B T \sin^2(\theta_{fg})} \left(P_{ext} + \frac{2\gamma_{se}}{r_f} \right) \left(\frac{3}{g_r} \right) \quad (61)$$

$$f(\theta) = 1 - 1.5 \cos(\theta) + 0.5 \cos^3(\theta) \quad (62)$$

Here, $r_f = 5 \times 10^{-7}$ m is the grain boundary bubble radius, $f(\theta_{fg})$ is a function accounting for the shape of the bubbles, $\theta_{fg} = 50^\circ$ is the semi-dihedral angle between bubble surfaces, $f_B = 0.5$ is the grain boundary fraction at saturation, k_B is the Boltzmann constant, T is the temperature in K, $\gamma_{se} = 0.626$ Jm⁻² is the surface tension of the bubbles, and g_r is the grain radius. The externally applied hydrostatic pressure is assumed to be zero ($P_{ext} = 0$) for simplicity.

3.6. Gap/plenum pressure calculation

The pressure in the gap and plenum is computed based on the ideal gas law,

$$P = \frac{nR}{\int_1^V \frac{1}{T} dV} \quad (63)$$

where P is the gap/plenum pressure, n is the moles of gas, R is the ideal gas constant, T is the temperature and V is the volume of the cavity. The gas volume is divided into sub-volumes, which are calculated individually and added together. The temperature T is calculated as the average temperature of the pellet exterior and cladding interior surface.

4. Results and discussions

In this section, the modeling results of enhanced thermal conductivity UO₂–BeO fuel behavior in CAMPUS are presented for a 2D axisymmetric LWR fuel rodlet. The CAMPUS code predictions are compared with those from BISON, ABAQUS and FRAPCON

(see Fig. 6) to verify and validate the CAMPUS code capability. The CAMPUS code is then subsequently applied to predict the behavior of enhanced thermal conductivity $\text{UO}_2\text{-BeO}$ fuels with different weight fractions of BeO, and the results obtained from these cases are discussed in the following section (see Figs. 7–13).

A 2D axisymmetric simplified fuel pellet was modeled with a typical finite element mesh as shown in Fig. 5(b). The geometry includes an individual $\text{UO}_2\text{-BeO}$ pellet, Zr-4 cladding, an initial 80 μm pellet-cladding gap, and computationally counted upper plenum. The plenum to fuel length ratio was set to 0.045. A uniform convective boundary at the cladding outer wall is used to simulate the heat transfer from the cladding to the outer flowing coolant. The linear power is assumed to rise linearly over three hours and then is held constant for 3.2 years. Typical PWR operating conditions were used as given in Table 3.

To verify and validate the CAMPUS code simulation capability, a comparison of the CAMPUS code predictions was performed for UO_2 fuel behavior against those from BISON, ABAQUS and FRAPCON codes. Fig. 6(a) shows the predicted temperatures at the fuel centerline, fuel outer surface and cladding inner surface during power up and steady state operation with four codes. The CAMPUS predicted temperature profiles are found to be consistent with those from the other codes. Some differences among the code predictions attribute to the models employed, for example, fuel densification and fission gas release models. As shown in Fig. 6(b), the CAMPUS predicted

Table 3

Input parameters for the axisymmetric problem.

Linear average power (W/cm)	200
Fast neutron flux ($\text{n/m}^2\text{ s}$)	9.5×10^{17}
Coolant pressure (MPa)	15.5
Coolant temperature (K)	530
Coolant convection coefficient ($\text{W/m}^2\text{ K}$)	7500
Rod fill gas	Helium
Fill gas initial pressure (MPa)	2.0
Initial fuel density	95% theoretical

gap size evolution is very similar to that from BISON and ABAQUS, while the FRAPCON predicted gap size decrease dramatically and results in an earlier gap closure due to a different mechanical model (fuel densification) used.

As shown in Fig. 6(c), the fission gas release from CAMPUS was earlier than from BISON, and the release fraction was a little bit larger than BISON, which further induced a higher plenum pressure as shown in Fig. 6(d), pushing the cladding outward and a delayed gap closure, owing to different fission gas release models used in BISON and CAMPUS. The fission gas release and plenum pressure, however, follow similar evolution trends by all the four codes.

The CAMPUS code was then subsequently applied to predict the behavior of enhanced thermal conductivity $\text{UO}_2\text{-BeO}$ fuels with different weight fractions of BeO. Fig. 7 shows the predicted

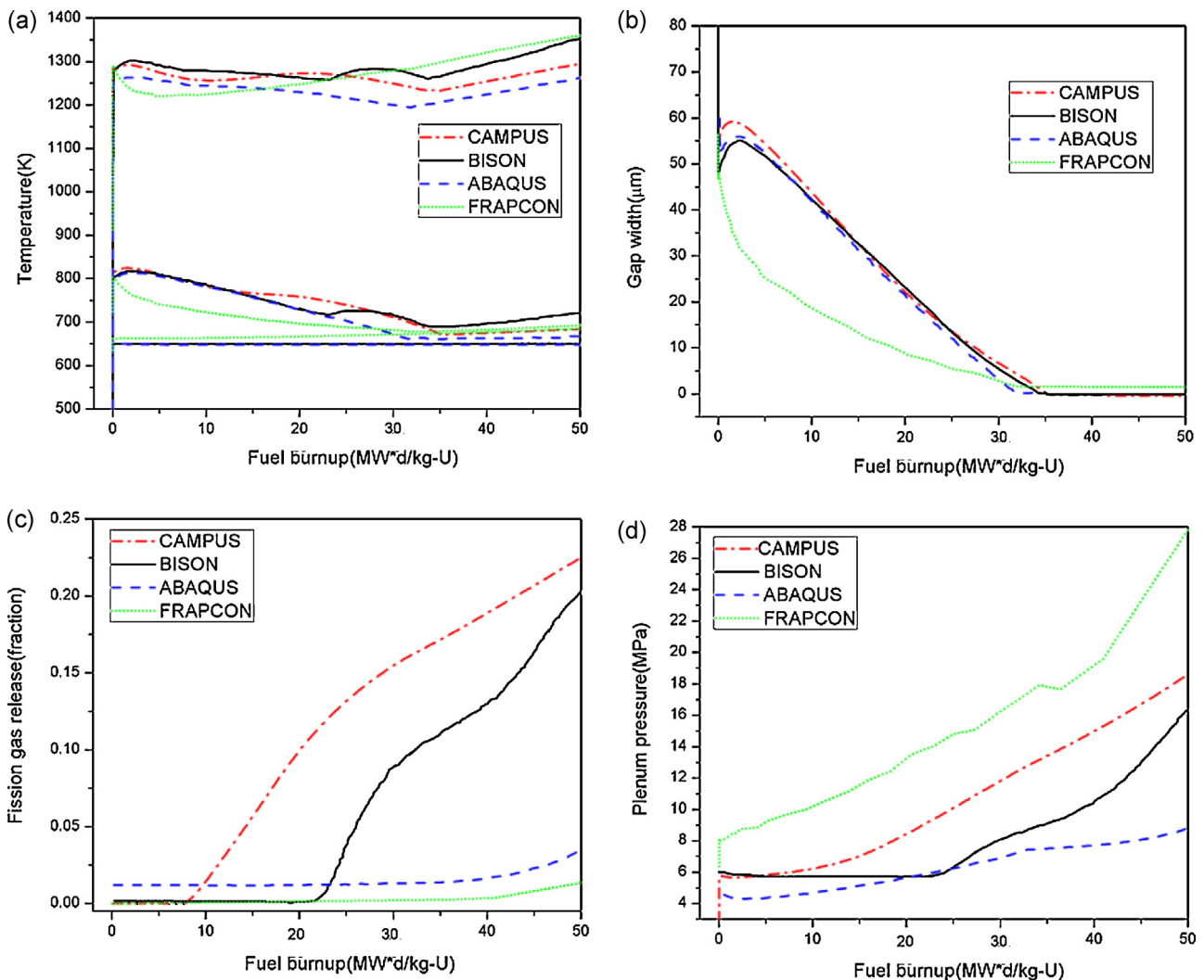


Fig. 6. Comparison of CAMPUS code to BISON, ABAQUS and FRAPCON. (a) Temperature profile; (b) Gap width; (c) Fission gas release; (d) Plenum pressure.

temperature evolutions by CAMPUS code at the fuel centerline, fuel outer surface and cladding inner surface during power up and steady operation. The fuel temperature is found to increase to a certain temperature rapidly during the initial power-up, then a drop in fuel temperature due to the creep-down of cladding and fuel swelling causing the gradual gap closure. When the gap is closed, the fuel temperature is found to increase because the fuel thermal conductivity decreases with the increase of fuel burnup.

From Fig. 7, we can see that the fuel temperature can be significantly lowered using the enhanced thermal conductivity $\text{UO}_2\text{-BeO}$ fuel. For the case of $\text{UO}_2\text{-13.6 wt\%BeO}$ fuel and zircaloy cladding in Fig. 7(d), the fuel centerline temperature was lowered about 350 K compared to the UO_2 fuel as shown in Fig. 7(a). Even for a small amount of BeO addition, for example, 0.6 wt% and 1.2 wt% in Fig. 7(b) and (c), the fuel centerline temperature can be lowered about 120–140 K compared to the UO_2 fuel. However, there is no significant difference for $\text{UO}_2\text{-BeO}$ fuel outer surface temperature and zircaloy cladding inner surface temperature compared to UO_2 fuel and zircaloy cladding as shown in Fig. 7.

The gap size evolution is shown in Fig. 8 for the four different cases, i.e., UO_2 fuel and zircaloy cladding; $\text{UO}_2\text{-0.6 wt\%BeO}$ fuel and zircaloy cladding; $\text{UO}_2\text{-1.2 wt\%BeO}$ fuel and zircaloy cladding; and $\text{UO}_2\text{-13.6 wt\%BeO}$ fuel and zircaloy cladding. We can see the initial gap size was about 80 μm , the gap size then decreased suddenly, which was caused by thermal expansion, there was then a short gap size increase period after the sudden decrease, which is caused by fuel densification, and at last the gap size was found to decrease

until the fuel and cladding contact. The gap closure is a combined effect of fuel swelling, thermal expansion and cladding creep down to the fuel. From Fig. 8, we can see the gap was closed earlier for UO_2 fuel/zircaloy cladding, then the gap was closed at later times for $\text{UO}_2\text{-0.6 wt\%BeO}$ fuel/zircaloy cladding and $\text{UO}_2\text{-1.2 wt\%BeO}$ fuel/zircaloy cladding, and at last the gap was closed for $\text{UO}_2\text{-BeO}$ fuel/zircaloy cladding. The BeO addition not only increases the fuel thermal conductivity, which lowers the fuel temperature, but also lowers the fuel density. According to Eqs. (51) and (52), with BeO addition, the fuel density decreases, and the volumetric swelling increases. So, the overall effect on fuel swelling is from both fuel temperature and density. However, the temperature plays a dominant role, and $\text{UO}_2\text{-BeO}$ fuel/zircaloy cladding results in a delay in gap closure as shown in Fig. 8.

The total gap conductance is shown in Fig. 9. The total conductance across the gap is the sum of gas conductance (h_{gas}), conductance due to solid–solid contact (h_{solid}) and the conductance due to radiant heat transfer (h_{rad}), which is a combined effect determined by the fuel outer surface and cladding inner surface temperatures, thermal conductivities of fuel, gas and cladding, contact pressure, surface conditions, related geometry changes and etc. The gap conductance first increases rapidly due to a significant temperature increase of the fuel outer surface and cladding inner surface resulting in a significant increase in radiative heat conductance according to Eq. (38), then slightly decreases due to the counter effect of the fuel thermal conductivity increase and cladding/gas thermal conductivity decrease with the burnup

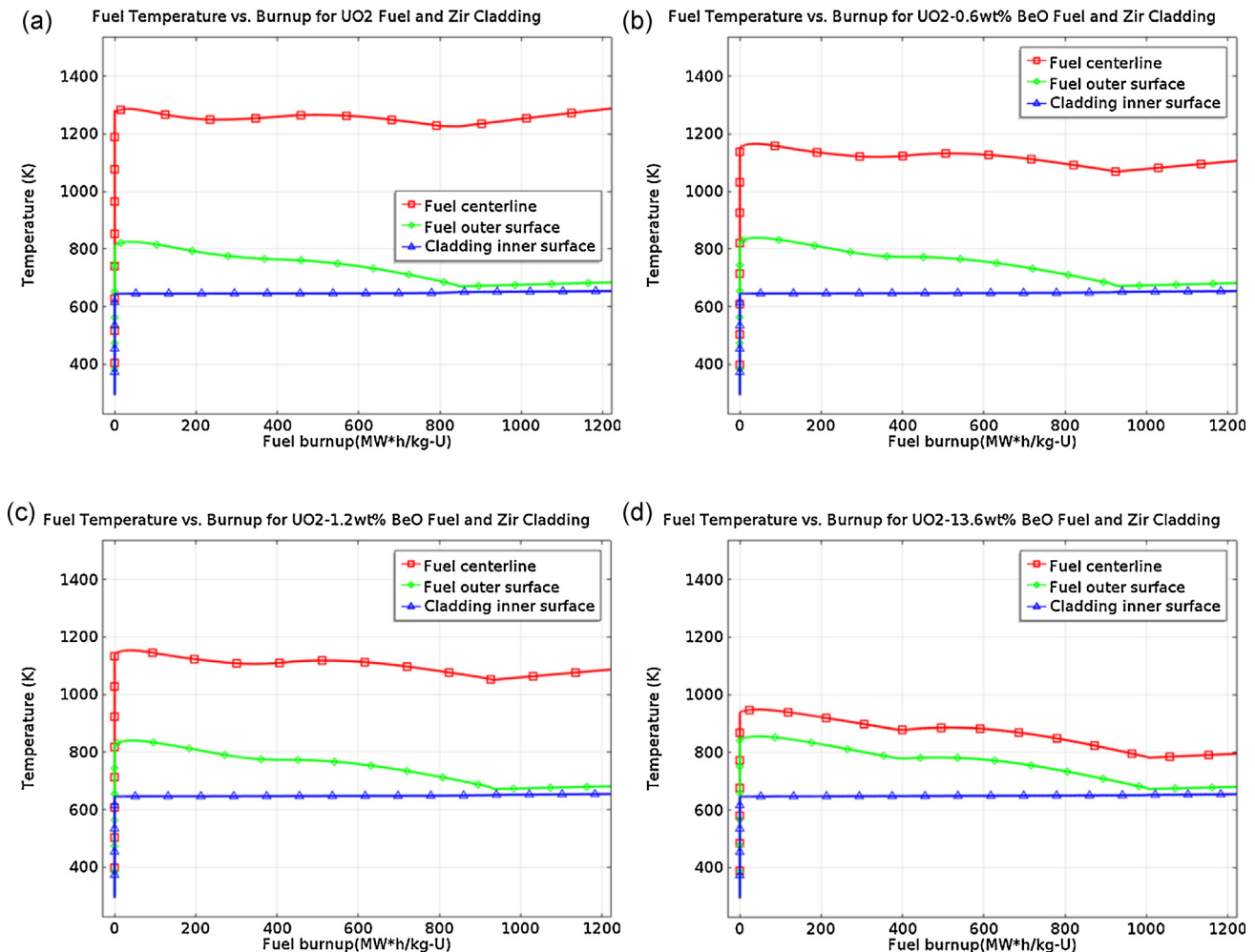


Fig. 7. Temperature at fuel centerline, outer surface and cladding inner surface with (a) UO_2 fuel and zircaloy cladding; (b) $\text{UO}_2\text{-0.6 wt\%BeO}$ fuel and zircaloy cladding; (c) $\text{UO}_2\text{-1.2 wt\%BeO}$ fuel and zircaloy cladding; (d) $\text{UO}_2\text{-13.6 wt\%BeO}$ fuel and zircaloy cladding.

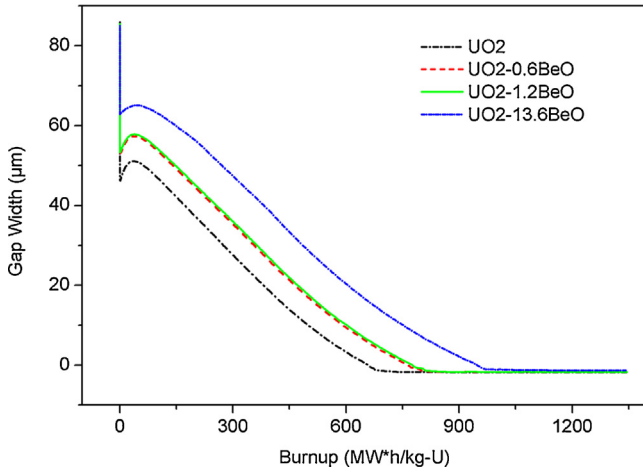


Fig. 8. Gap size evolution with different type fuels and zircaloy cladding.

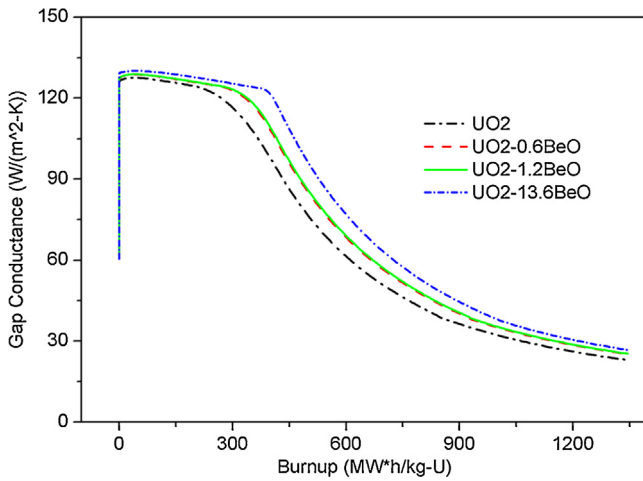


Fig. 9. Gap conductance evolution with different type fuels and zircaloy cladding.

increase, until a burnup of around 450–500 MW h/kg-U, and after that, the fuel thermal conductivity increase dominates resulting in a total gap conductance decrease. All the gap conductance evolutions follow a similar trend for the four cases with the lowest total gap conductance for UO₂ fuel/zircaloy cladding, and the highest total gap conductance for UO₂–13.6 wt%BeO fuel/zircaloy cladding.

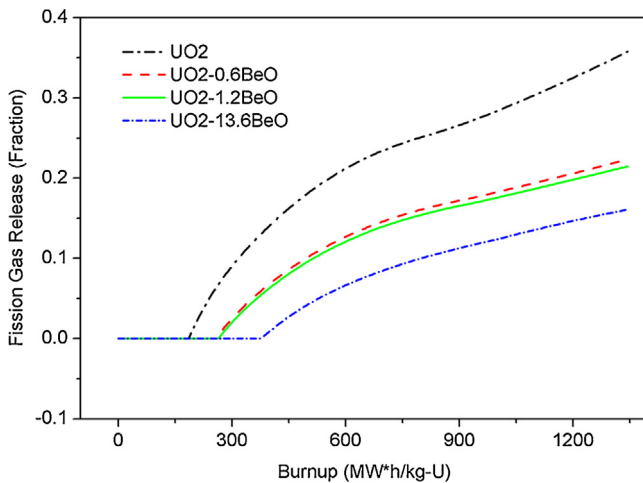


Fig. 10. Fission gas release evolution with different type fuels and zircaloy cladding.

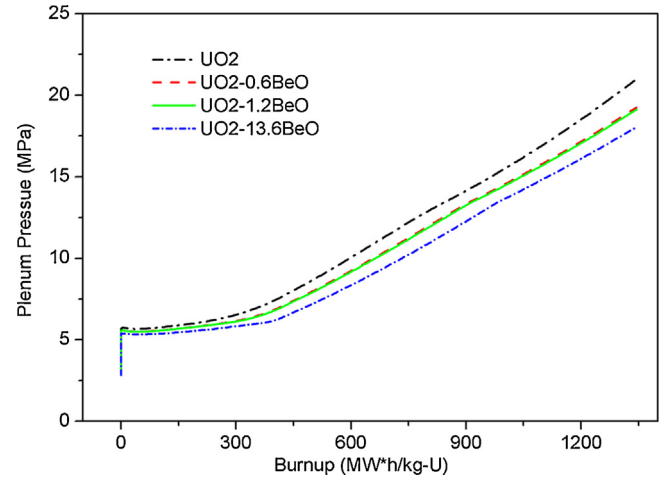


Fig. 11. Plenum pressure evolution with different type fuels and zircaloy cladding.

The fission gas release are depicted against fuel burnup in Fig. 10 for the four different cases, i.e., UO₂ fuel and zircaloy cladding; UO₂–0.6 wt%BeO fuel and zircaloy cladding; UO₂–1.2 wt%BeO fuel and zircaloy cladding; and UO₂–13.6 wt%BeO fuel and zircaloy cladding. The UO₂ case experienced the highest fission gas release fraction as expected due to its low thermal conductivity and high fuel temperature. The fission gas release can be reduced significantly by 55% at the fuel burnup of about 1200 MW h/kg-U for UO₂–13.6 wt%BeO fuel and zircaloy cladding as shown in Fig. 10. Even for small fractions of BeO addition, for example, 0.6 wt% BeO and 1.2 wt% BeO, the fission gas release can be reduced by 37% and 39%, respectively, at the fuel burnup of about 1200 MW h/kg-U. From Fig. 10, we can also see the earliest start time to release fission gas was found to be the case of UO₂ fuel and zircaloy cladding, which can also conclude that fission gas release is dominated by temperature.

Fig. 11 shows the gap/plenum pressure evolution against fuel burnup for the four different cases. UO₂ fuel had highest gap/plenum pressure since its lower thermal conductivity leads to higher fuel temperature, more fission gas release and at last higher pressure. On the other hand, UO₂–13.6 wt%BeO fuel and zircaloy cladding experienced lowest gap/plenum pressure as shown in Fig. 11.

Fig. 12 shows the evolution of the oxygen deviation from the stoichiometry, which is designated as x :

$$x = O/U - 2 \quad (64)$$

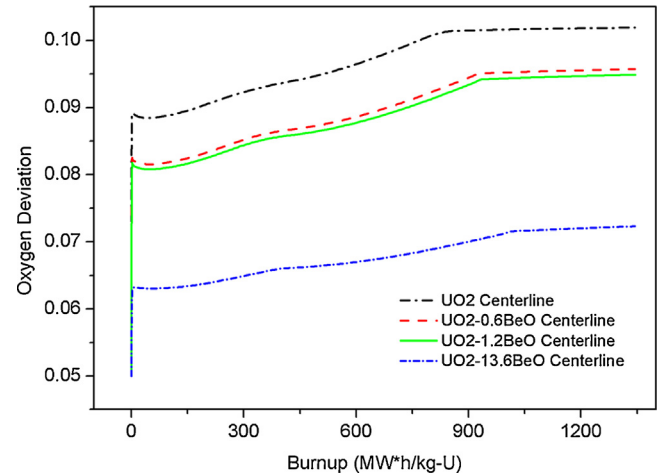


Fig. 12. Oxygen deviation evolution of fuel centerline and outer surface with different type fuels and zircaloy cladding.

For the light water reactor, the fuel is in the hyperstoichiometric state, i.e., $x > 0$. We can see the oxygen redistribution is more significant in the UO_2 fuel compared to the UO_2 -BeO composite fuels due to the relatively high fuel temperature of the UO_2 fuel.

The radial displacements are shown in Fig. 13. From Fig. 13(a)–(d), we can see that the fuel displacements decrease with the BeO fraction increase from 0 wt% to 13.6 wt%, which leads to the fuel temperature decrease, at the same burnup. Fig. 13(d)–(f) show

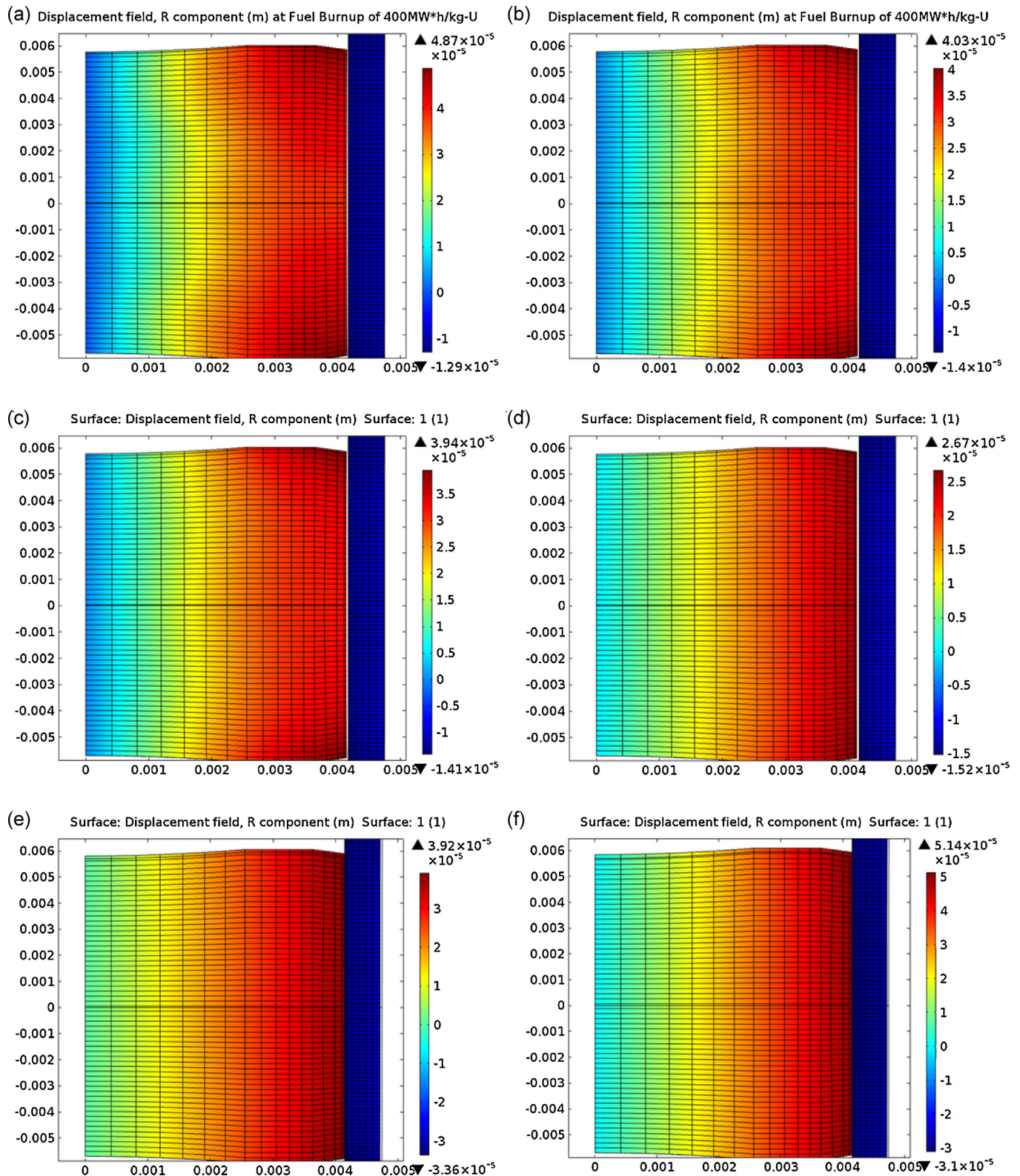


Fig. 13. Fuel and cladding radial displacement with different type fuels and zircaloy cladding at different fuel burnup, all the distances are in unit meter. (a) UO_2 displacement at 400 MW h/kg-U; (b) UO_2 -0.6 wt% BeO displacement at 400 MW h/kg-U; (c) UO_2 -1.2 wt% BeO displacement at 400 MW h/kg-U; (d) UO_2 -13.6 wt% BeO displacement at 400 MW h/kg-U; (e) UO_2 -13.6 wt% BeO displacement at 800 MW h/kg-U; (f) UO_2 -13.6 wt% BeO displacement at 1200 MW h/kg-U.

the fuel displacements increase with the fuel burnup increase at a fixed BeO fraction in the composite fuel.

5. Conclusions

In summary, the enhanced thermal conductivity UO_2 –BeO fuel performance has been modeled in a light water reactor. Constitutive laws currently implemented include UO_2 –BeO temperature and burnup dependent thermal properties, solid and gaseous fission product swelling, fuel densification, fuel thermal and irradiation creep, fission gas generation, diffusion and release, cladding thermal and irradiation creep, fuel-clad gap mechanics (contact and gap heat transfer), and gap/plenum gas behavior during irradiation. Through fully coupled multiphysics modeling, a more complete and detailed understanding was achieved for enhanced thermal conductivity UO_2 –BeO fuel performance in a light water reactor, which has never been done before according to the authors' knowledge.

BeO has high melting point, low neutron absorption, low thermal expansion coefficient and chemical compatibility with current UO_2 fuels which makes it a suitable material for thermal conductivity enhancement. In our modeling and simulation work, the UO_2 –BeO enhanced thermal conductivity nuclear fuel showed the capability of decreasing the fuel temperatures, increasing the gap total conductance, reducing the fission gas release, decreasing the gap/plenum pressure, lessening oxygen redistribution, and facilitating a reduction in pellet cladding interaction through lessening thermal stresses that result in fuel cracking, relocation, and swelling. The safety of the reactor operation would be then improved.

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